Chapter 3: Cascading Style Sheets (CSS)

What students will learn

Why CSS is used with HTML

CSS rules, selectors, properties, and values

Type, universal, class, and ID selectors

span and div elements

How CSS rules cascade and which rule wins

Style attribute, style element, and external CSS files

Colors, fonts, text, borders, padding, and margins

Element box and common CSS length units

How the chapter combines these ideas in a case study

3.1–3.2 CSS: Content vs. Appearance

HTML describes the content of a web page.

CSS describes how that content looks.

CSS can control color, text, fonts, spacing, borders, and more.

Keeping CSS separate from HTML makes changes easier.

Simple idea: HTML = what is on the page; CSS = how it looks.

3.3 CSS Rules

A CSS rule tells the browser how selected HTML elements should look.

A selector tells the browser which elements to match.

A property tells what part of the appearance to change.

A value tells the browser what setting to use.

```
hr {width: 50%;}
```

Here: hr is the selector, width is the property, and 50% is the value.

Type Selector and Universal Selector

Type selector: uses an HTML element name, such as hr or p.

Universal selector: uses * and matches every element.

Use a comma-separated list when the same rule should apply to several element types.

```
hr {width: 50%;}

* {text-align: center;}

h2, p {font-style: italic; color: blue;}
```

3.4 Example: Tree Poem

The book's Tree Poem example uses three rules.

All text is centered with the universal selector.

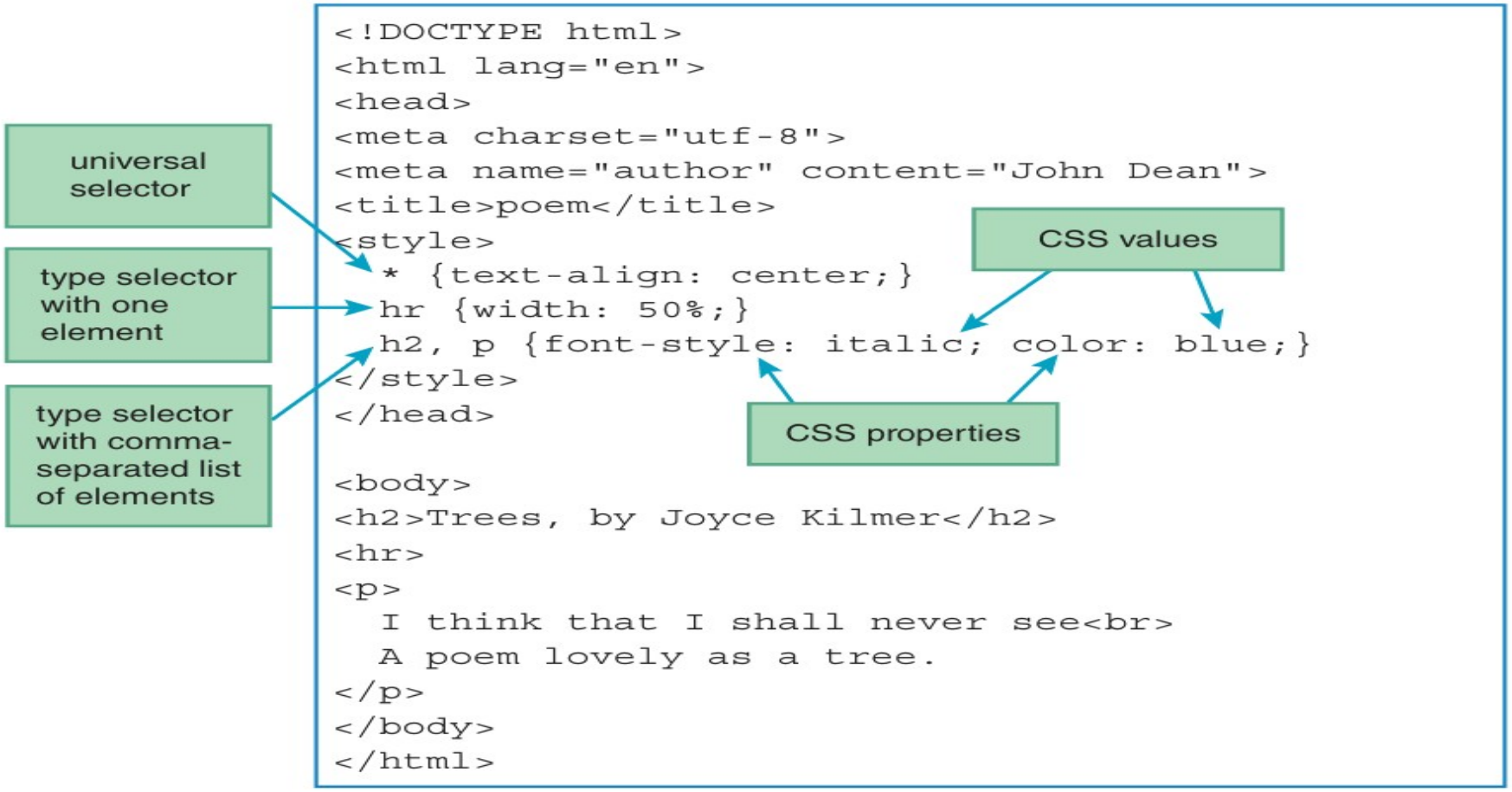
The horizontal line is made 50% of its container width.

The heading and paragraph are italic and blue.

```
<style>
  * {text-align: center;}
  hr {width: 50%;}
  h2, p {font-style: italic; color: blue;}
</style>
```

Teaching point: one CSS rule can format more than one element type.

3.4 Example: Tree Poem



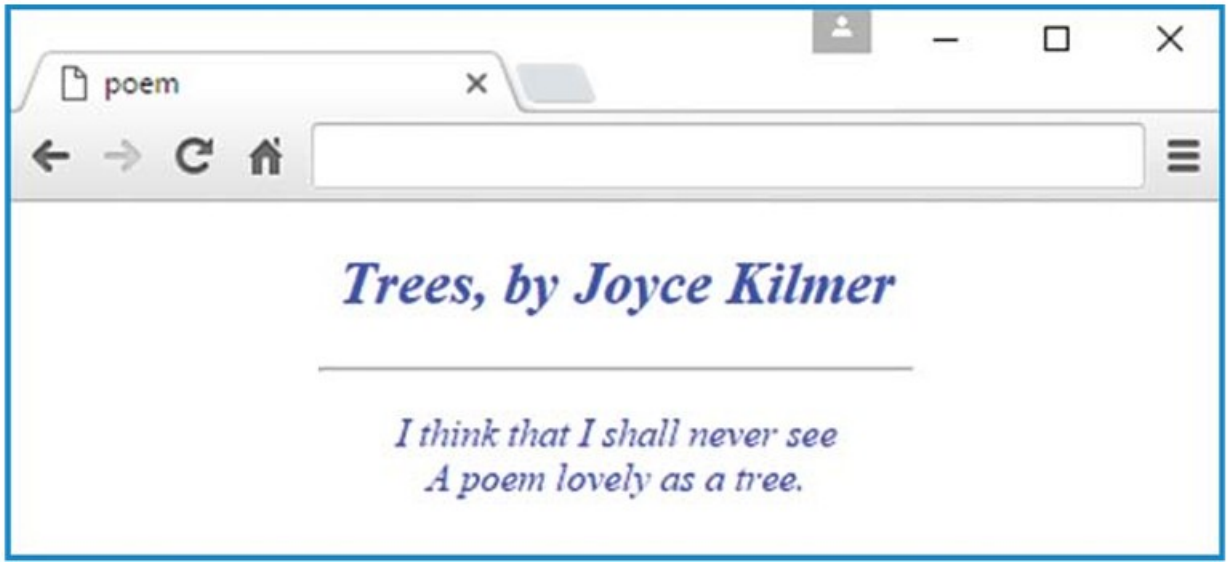

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="author" content="John Dean">
<title>poem</title>
<style>
  * {text-align: center;}
  hr {width: 50%;}
  h2, p {font-style: italic; color: blue;}
</style>
</head>

<body>
<h2>Trees, by Joyce Kilmer</h2>
<hr>
<p>
   I think that I shall never see<br>
   A poem lovely as a tree.
</p>
</body>
</html>
```

FIGURE 3.1 Source code for Tree Poem web page

Teaching point: one CSS rule can format more than one element type.

3.4 Example: Tree Poem



3.5 CSS Syntax and Style

Put the style element in the head section.

Short rule: selector {property: value;}

Put a semicolon after each property-value pair.

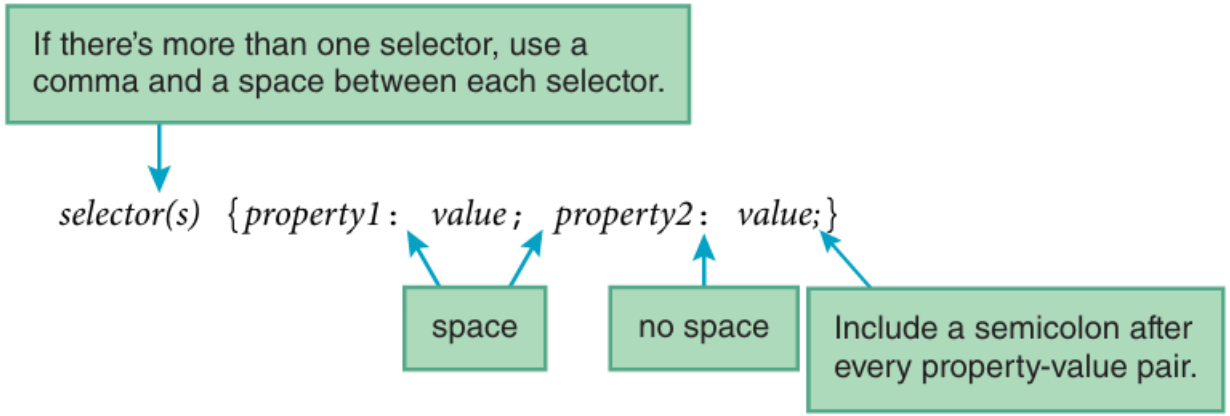
Use a comma and a space between multiple selectors.

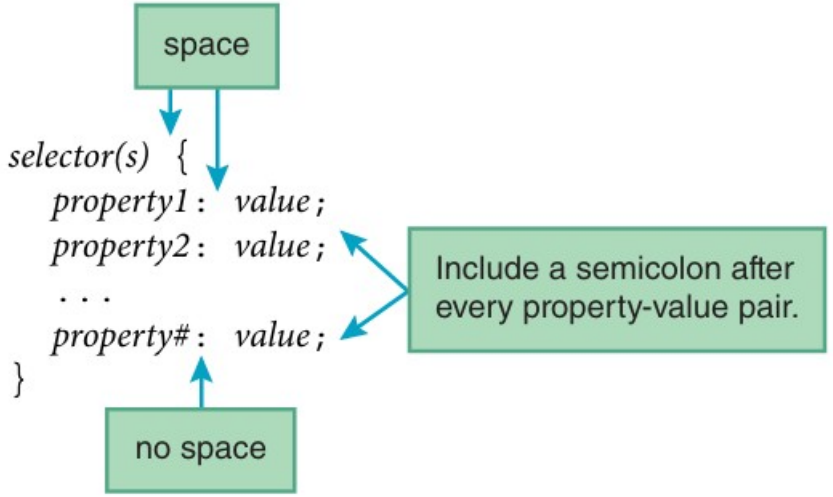
For longer rules, put each property on its own line.

Following a common style makes code easier to read and maintain.

```
p {
  color: blue;
  text-align: center;
}
```

3.5 CSS Syntax and Style





3.6 Class Selectors

A class selector starts with a dot: .class-name

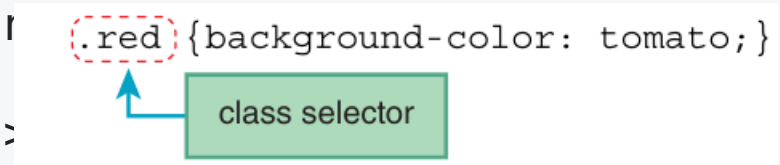
It matches elements whose class attribute has that value.

A class value does not have to be unique.

Choose a class name that describes the purpose.

```
.red {backgrou
```

```
<q class="red">
```

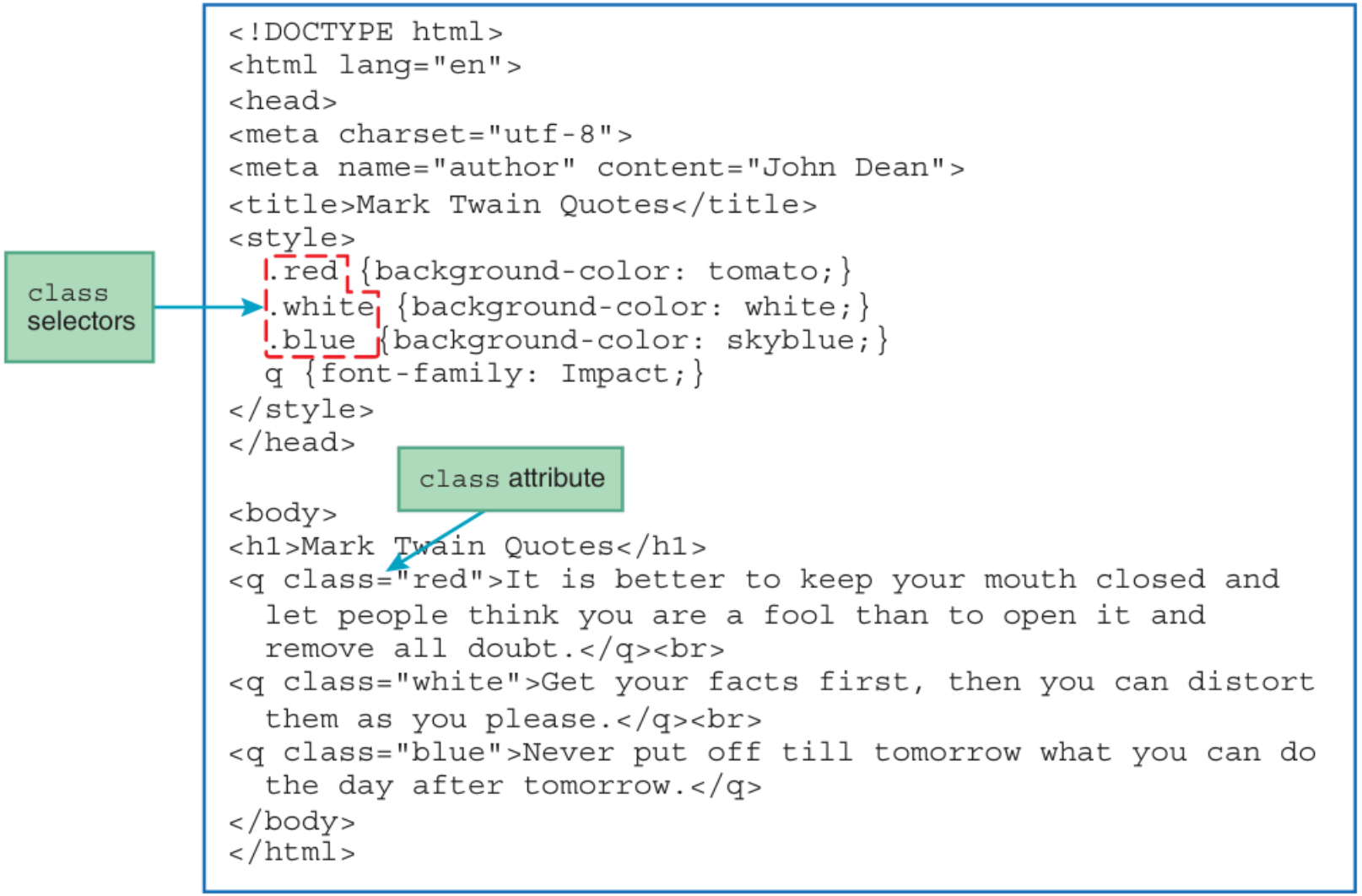


3.6 Class Selectors

With a Class selector we can have more than one class separated by commas sharing one CSS Rule

```
.red, .conspicuous, h1 {background-color: tomato;}
```

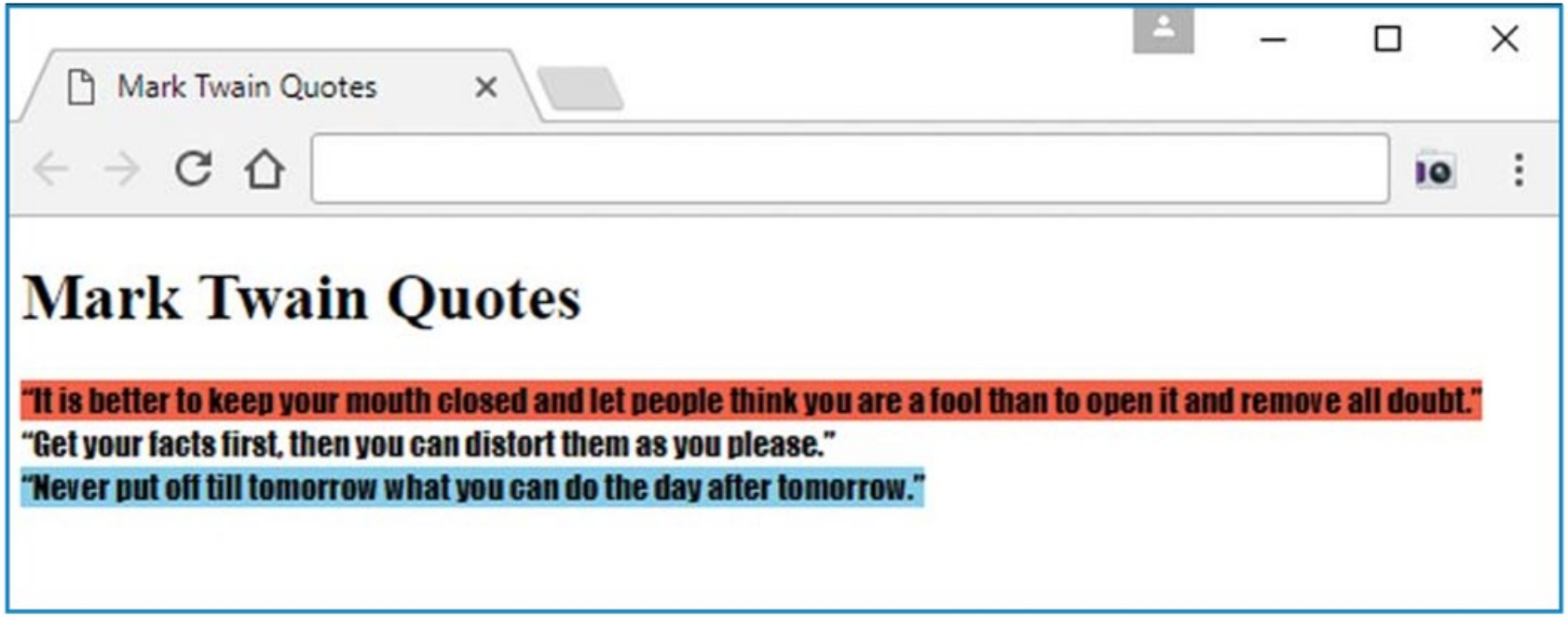
3.6 Class Selectors




```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="author" content="John Dean">
<title>Mark Twain Quotes</title>
<style>
  .red {background-color: tomato;}
  .white {background-color: white;}
  .blue {background-color: skyblue;}
  q {font-family: Impact;}
</style>
</head>

<body>
<h1>Mark Twain Quotes</h1>
<q class="red">It is better to keep your mouth closed and
  let people think you are a fool than to open it and
  remove all doubt.</q><br>
<q class="white">Get your facts first, then you can distort
  them as you please.</q><br>
<q class="blue">Never put off till tomorrow what you can do
  the day after tomorrow.</q>
</body>
</html>
```

3.6 Class Selectors



Class Selector with Element Prefix

q.blue matches only q elements whose class is blue.

h1.blue matches only h1 elements whose class is blue.

This is more specific than simply using .blue.

It can make CSS easier to maintain when different element types need different styles.

```
q.blue {background-color: skyblue;}
h1.blue {color: blue;}
```

Class Selectors with * prefixes

you can preface a class selector with an *

is the universal selector

it matches all elements

what would the following CSS rule do?

```
*.class-value {property1: value; property2: value;}
```

```
*.big-warning {font-size: x-large; color:
red;}
```

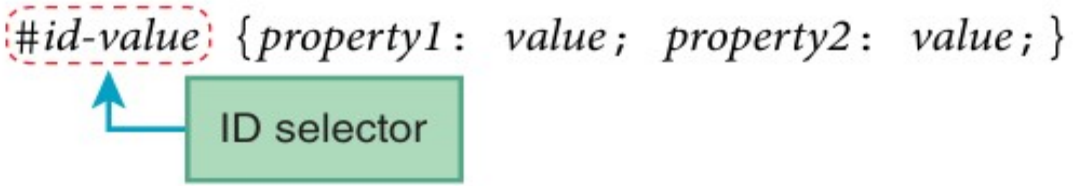
3.7 ID Selectors

An ID selector starts with #.

It matches an element through its id attribute.

An id value must be unique within a web page.

ID selectors are useful when exactly one element should be targeted.




```
#lizards-lounge {color: green;}

<h3 id="lizards-lounge">Lizards Lounge</h3>
```

Class vs. ID

Class: can be used by many elements.

ID: should identify one unique element on the page.

Class selector uses .

ID selector uses #.

```
.warning {color: red;}
#main-heading {color: green;}
```

CSS Combinators

A combinator is something that defines the relationship between two or more selectors.
A CSS selector can contain more than one selector.
Between the selectors, we can include a combinator, to create a more specific selection.
There are four different combinators in CSS:

- Descendant combinator (space)
- Child combinator (>)
- Next sibling combinator (+)
- Subsequent-sibling combinator (~)

```
.warning {color: red;}
#main-heading {color: green;}
```

CSS Combinators

Descendant Combinators
The descendant combinator matches all elements that are descendants (children, grandchildren, etc.) of a specified element.

```
div p {
  background-color: yellow;
}
```

selects all <p> elements inside elements

Child Combinator (>)
The child combinator selects all elements that are direct children of a specified element.

```
div > p {
  background-color: yellow;
}
```

elects all <p> elements that are direct children of

CSS Combinators

The next sibling combinator is used to select an element that is directly after a specific element. Sibling elements must have the same parent element.

```
div + p {
  background-color: yellow;
}
```

selects the first <p> element that immediately follows a , and share the same parent

CSS-Pseudo classes

A pseudo-class is used to define a special state of an element.
For example, it can be used to:
Style an element when a user mouse over it
Style visited and unvisited links differently
Style an element when it gets focus
Pseudo-classes are always denoted by a single colon (:) followed by the pseudo-class name:

```
selector:pseudo-class-name {
  CSS properties
}
```

```
.submit-btn:hover {
      background-color:
aquamarine;
}
```

submit Form

CSS-Pseudo classes

A pseudo-class is used to define a special state of an element.
For example, it can be used to:
Style an element when a user mouse over it
Style visited and unvisited links differently
Style an element when it gets focus
Pseudo-classes are always denoted by a single colon (:) followed by the pseudo-class name:

```
selector:pseudo-class-name {
  CSS properties
}
```

```
.submit-btn:hover {
      background-color:
aquamarine;
}
```

submit Form

Pseudo-Elements

A CSS pseudo-element is used to style specified parts of an element.
For example, it can be used to:
Style the first letter, or line, of an element
Insert content before, or after, the content of an element

Pseudo-elements are denoted by a double colon (::) followed by the pseudo-element name:

```
p::first-letter {
      font-size:20pt;
      color: #ff0000;
      font-variant: small-caps;
}
```

```
selector::pseudo-element-name {
  CSS properties
}
```

You can use the ::first-line pseudo-element to add a special effect to the first line of a text. Some more text. And even more, and more, and more, and more, and more, and more, and more, and more, and more, and more, and more, and more.

3.8 span and div

Use span when the styled text is inside surrounding text.

Use div when the content should act like a block.

span and div are generic elements; by themselves they do not add special meaning.

They are useful as containers for CSS.

```
<div class="orange">Text and more text...</div>

Price: <span class="white orange-background">$10</span>
```

3.9 Cascading

CSS can have rules in different places.

When rules conflict, the browser uses the rule with higher priority.

More specific rules can override more general rules.

The chapter presents these places from highest to lowest priority:

1. style attribute
2. style element
3. external CSS file
4. user browser settings
5. browser default settings

3.10 style Attribute and style Element

style attribute puts CSS directly on one HTML element.

It has the highest priority in the chapter's list.

The book advises avoiding it when possible because it mixes presentation with content.

The style element holds rules for the page and is normally placed in head.

```
<h2 style="text-decoration:underline;">Welcome!</h2>
```

Use a style element or external CSS when the same formatting is needed in many places.

3.11 External CSS Files

Put CSS rules in a separate file ending in .css.

Connect the HTML page to that file with link.

This keeps large HTML files easier to understand.

One external CSS file can provide common styles for several pages.

```
<link rel="stylesheet" href="pumpkinPatch.css">
```

External CSS Example

The Pumpkin Patch example moves the same rules out of the HTML file.

The HTML head contains the link element.

The CSS file contains only CSS rules and comments.

```
/* John Dean */
.orange {color: darkorange;}
.white {color: white;}
.black {color: black;}
.orange-background {background-color: orange;}
```